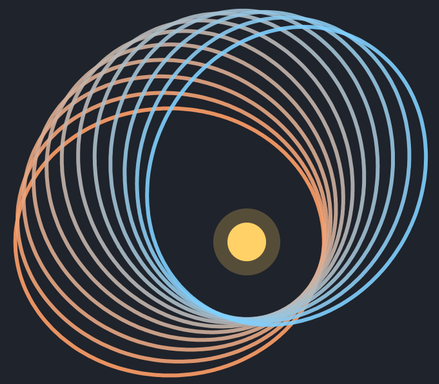
Pi & Planck

GENERAL RELATIVITY

Einstein and the Orbit of Mercury

Newton's gravity left 43 arcseconds per century of Mercury's motion unexplained. Einstein's answer came on 18 November 1915.

THE ORBIT THAT TURNS



Each orbit is an ellipse, but its closest point (perihelion) creeps forward. The shift is drawn exaggerated.

THE 43 ARCSECONDS

Observed advance of the perihelion: 5600″ per century

5557″ explained by Newton: other planets, the Sun's shape, Earth's moving frame

43″ left over

EINSTEIN'S FORMULA

$$\Delta\varphi = \frac{6\pi GM}{c^2\, a\,(1-e^2)}$$

advance per orbit, in radians

a = 57.91 million km · e = 0.2056 · period 87.97 days

0.104″ per orbit × **415.2** orbits per century = **43.0″** per century

HISTORY

1859: Le Verrier

The French astronomer found that Mercury's perihelion advanced faster than Newtonian gravity allowed.

18 Nov 1915: Einstein

He presented to the Prussian Academy that general relativity gives the missing advance, about 43″ per century.

Today: 42.98 ± 0.04″

Modern measurements agree with the prediction to about one part in a thousand.

43″ per century: the part of Mercury's perihelion advance Newton could not explain

0.104″ advance in a single 88-day orbit, about 1/35 000 of a degree

3 million years for the orbit to turn once all the way round

Sources: A. Einstein, Sitzungsberichte der Preussischen Akademie der Wissenschaften (1915) 831–839 · U. Le Verrier (1859) · Tests of General Relativity: A Review, arXiv:1705.04397 · Orbit values from NASA planetary fact sheet; calculation done for this graphic